

# SLIDES WEEK 21

## ALGEBRAIC STRUCTURES

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# WEEK 21: GOALS

## Last time:

- New algebraic structures: Fields!
- Subfields
- Quick Subfield Theorem
- Homomorphism of rings, ID and fields

## Today:

- Isomorphisms of rings, ID and fields
- Ideals

# REMINDER: RING HOMOMORPHISM, ID HOMOMORPHISMS AND FIELD HOMOMORPHISMS

## Definition

Let  $R$  and  $S$  be rings. A **ring homomorphism** from  $R$  to  $S$  is a mapping

$$\theta : R \rightarrow S$$

that satisfies:

- $\theta(a + b) = \theta(a) + \theta(b)$ , and
- $\theta(ab) = \theta(a)\theta(b)$ ,

for all  $a, b \in R$ .

# RING, INTEGRAL DOMAIN AND FIELD HOMOMORPHISMS

## Remark

Recall that

$$\text{Fields} \subset \text{Integral Domains} \subset \text{Rings}.$$

Thus

- an **integral domain homomorphism** is just a **ring homomorphism** between two integral domains.
- Similarly, a **field homomorphism** is just a **ring homomorphism** between two fields.

# EXAMPLE: COMPLEX CONJUGATION

## Example

**Complex conjugation map**  $f : \mathbb{C} \rightarrow \mathbb{C}$  given by

$$f(a + ib) = a - ib.$$

Let us show that this is a ring homomorphism.

- Conjugation preserves sum:

$$\begin{aligned} f((a + ib) + (c + id)) &= f((a + c) + i(b + d)) \\ &= (a + c) - i(b + d) \\ &= (a - ib) + (c - id) \\ &= f(a + ib) + f(c + id). \end{aligned}$$

# EXAMPLE: COMPLEX CONJUGATION - PART 2

## Example

- Conjugation preserves multiplication:

$$\begin{aligned} f((a + ib) \cdot (c + id)) &= f((ac - bd) + i(ad + bc)) \\ &= (ac - bd) - i(ad + bc) \\ &= (a - ib) \cdot (c - id) \\ &= f(a + ib) \cdot f(c + id). \end{aligned}$$

## Remark

- Since  $\mathbb{C}$  is an integral domain, the map  $f$  is also an **integral domain homomorphism**.
- Since  $\mathbb{C}$  is a field, the map  $f$  is also a **field homomorphism**.

# RING ISOMORPHISMS



# RING ISOMORPHISMS

## Definition.

Let  $R$  and  $S$  be rings. An **isomorphism** from  $R$  to  $S$  is a mapping

$$\theta : R \rightarrow S$$

$$\forall s \in S, \exists r \in R \text{ such that } \theta(r) = s$$

that is **injective**, **surjective**, and a **ring homomorphism**.

$$\theta(x + y) = \theta(x) + \theta(y) \text{ and } \theta(xy) = \theta(x)\theta(y)$$

$$\theta(x) = \theta(y) \implies x = y$$

## Definition.

Let  $R$  and  $S$  be rings. An **isomorphism** from  $R$  to  $S$  is a mapping

$$\theta : R \rightarrow S$$

that is **injective**, **surjective**, and a **ring homomorphism**.

If such isomorphism exist we write  $R \cong S$ , and say that  $R$  and  $S$  are **isomorphic**.

# EXAMPLE 1

## Example

**Today:** The complex conjugation map  $f : \mathbb{C} \rightarrow \mathbb{C}$  given by  $f(a + ib) = a - ib$  is a ring homomorphism.

Let us check whether  $f$  is a ring **isomorphism**.

■ **Injective:** Suppose  $f(a + ib) = f(c + id)$ , we must show

$$a + ib = c + id.$$

In fact,  $f(a + ib) = f(c + id)$  is equivalent to  $a - ib = c - id$ .

By comparing real and imaginary parts, we deduce that  $a = c$  and  $b = d$ .

In particular,  $a + ib = c + id$ .

## EXAMPLE 1 - PART 2

### Example (continuation)

- **Surjective:** Given  $a + ib \in C$ , we must find  $x \in \mathbb{C}$  such that

$$f(x) = a + ib.$$

Note that  $x = a - ib$  is the required element.

Consequently,  $f$  is a **ring isomorphism**.

### Remark

- Since  $\mathbb{C}$  is an integral domain, the map  $f$  is also an **integral domain isomorphism**.
- Since  $\mathbb{C}$  is a field, the map  $f$  is also a **field isomorphism**.

## EXAMPLE 2

### Example

**Last week:** The map  $f : \mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$  given by

$$f(x) = \overline{4}x, \quad \text{for all } x \in \mathbb{Z}/6\mathbb{Z}$$

is a ring homomorphism.

Let us check whether  $f$  is a ring isomorphism.

**Check injectivity:** If  $f(x) = f(y)$ , we must check whether  $x = y$ .

Now,  $f(x) = f(y)$  implies  $\overline{4}x = \overline{4}y$ .

Because  $\mathbb{Z}/6\mathbb{Z}$  is **not an integral domain**, cancellation laws **do not hold!** For instance, we have  $\overline{4} \cdot \overline{4} = \overline{16} = \overline{4}$  and  $\overline{4} \cdot \overline{1} = \overline{4}$ .

Thus,  $f(\overline{4}) = f(\overline{1})$ . We conclude  $f$  is **not** a ring isomorphism.

## Remark

- The map from the previous example is **not** surjective.
- Here are two ways to see this:
  - For finite sets, a map  $f : \mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$  is surjective if and only if it is injective.

Since  $f$  is not injective, it cannot be surjective.

- Using the definition: For  $f$  to be surjective, every  $y \in \mathbb{Z}/6\mathbb{Z}$  must have an  $x \in \mathbb{Z}/6\mathbb{Z}$  such that  $f(x) = y$ .

However,

$$\begin{aligned} f(\bar{0}) &= \bar{0}, & f(\bar{1}) &= \bar{4}, & f(\bar{2}) &= \bar{2}, \\ f(\bar{3}) &= \bar{0}, & f(\bar{4}) &= \bar{4}, & f(\bar{5}) &= \bar{2} \end{aligned}$$

So, there is no  $x \in \mathbb{Z}/6\mathbb{Z}$  such that  $f(x) = \bar{1}$ ,  $\bar{3}$ , or  $\bar{5}$ .

# ISOMORPHIC VS NON-ISOMORPHIC RINGS

## Isomorphic vs non-isomorphic rings

- Rings  $R$  and  $S$  are isomorphic if there is a ring isomorphism  $\varphi : R \rightarrow S$ .
- They are **not** isomorphic if no such isomorphism exists.

## Proving isomorphism vs non-isomorphism

- To show  $R \cong S$ , find a ring isomorphism  $\varphi : R \rightarrow S$ .
- To show  $R \not\cong S$ , prove no such isomorphism exists.
- Checking all possible maps  $\varphi : R \rightarrow S$  is too much work!
- Instead, use ring theoretical properties.

# NON-ISOMORPHIC RINGS

## Recipe for proving rings that are not isomorphic

- Find some property that one ring has, but that the other ring does not.
- Conclude they cannot be isomorphic.

## Examples

- $\mathbb{Z}$  and  $\mathbb{Z}_n$  are rings. However,  $\mathbb{Z}$  is infinite and  $\mathbb{Z}_n$  is finite. Thus, there is no isomorphism between them.
- $\mathbb{Z}$  and  $\mathbb{Q}$  are not isomorphic because  $\mathbb{Q}$  is a field, but  $\mathbb{Z}$  is not.
- $\mathbb{Z}_3$  and  $\mathbb{Z}_8$  are rings. However,  $\mathbb{Z}_3$  is a field, while  $\mathbb{Z}_8$  is not even an integral domain. Thus, they are not isomorphic.
- $\mathbb{Z}$  and  $2\mathbb{Z}$  are both rings. However,  $\mathbb{Z}$  has unity and  $2\mathbb{Z}$  has not. Thus, they are not isomorphic.



## Properties preserved by ring isomorphisms

If one of two **isomorphic rings**

- is **commutative**, then so is the other,
- **has unit**, then so does the other,
- **has a zero divisor**, then so does the other,
- is an **integral domain**, then so is the other,
- is a **field**, then so is the other.

# IDEALS

## Definition.

A subring  $I$  of a ring  $R$  is an **ideal** of  $R$  if

$$ar \in I \text{ and } ra \in I$$

for all  $a \in I$  and all  $r \in R$ .

We write  $I \triangleleft R$  to indicate that  $I$  is an ideal of  $R$ .

## Example

We know that  $2\mathbb{Z} \subset \mathbb{Z}$  is a subring. Now, given  $2z \in 2\mathbb{Z}$  and  $x \in \mathbb{Z}$ , we have

$$2z \cdot x = 2(zx) \in 2\mathbb{Z} \quad \text{and} \quad x \cdot 2z = 2(xz) \in 2\mathbb{Z}$$

because  $\mathbb{Z}$  and  $2\mathbb{Z}$  are commutative rings. Thus,  $2\mathbb{Z} \triangleleft \mathbb{Z}$ .

# EXAMPLES AND NON-EXAMPLES

## Non-example

Although  $\mathbb{Z}$  is a subring of  $\mathbb{Q}$ , it is **not an ideal**: setting  $a = 1$  and  $r = 1/2$  yields

$$ar = 1/2 \notin \mathbb{Z}.$$

## The trivial examples

For every ring  $R$ :

- $\{0\} \triangleleft R$  because  $0r = r0 = 0$ , for all  $r \in R$ ,
- $R \triangleleft R$  because  $R$  is closed with respect to its multiplication.

# IMPORTANT EXAMPLE: POLYNOMIAL RING

Recall:  $\mathbb{R}[x]$

Recall that  $\mathbb{R}[x]$  is the ring of all polynomials

$$a_0 + a_1x + \cdots + a_nx^n, \text{ with } a_i \in \mathbb{R} \text{ and } n \in \mathbb{N} \cup \{0\}.$$

Example.

The following subset of  $\mathbb{R}[x]$  is an ideal:

$$\mathcal{F} = \{(x^2 + 1)f(x) \mid f(x) \in \mathbb{R}[x]\}.$$

(Notice that  $\mathcal{F}$  is the set of all polynomials in  $\mathbb{R}[x]$  that are divisible by  $x^2 + 1$ .)

## EXAMPLE: POLYNOMIAL RING—PART 2

Let us show  $\mathcal{F} \triangleleft \mathbb{R}[x]$

One can show  $\mathcal{F}$  is a subring of  $\mathbb{R}[x]$  (Exercise!).

Given  $a = (x^2 + 1)f(x) \in \mathcal{F}$  and  $r = g(x) \in \mathbb{R}[x]$ , we must show

$$ar, ra \in \mathcal{F}.$$

We have

$$ar = ((x^2 + 1)f(x))g(x) = (x^2 + 1)(fg)(x) \in \mathcal{F}.$$

Similary,  $ra \in \mathcal{F}$ .

## Image & Kernel

Given a **ring homomorphism**  $\theta : R \rightarrow S$ , the **image** of  $\theta$  is

$$\text{Im}(\theta) = \{\theta(r) \mid r \in R\} \subset S,$$

and the **kernel** of  $\theta$  is

$$\text{Ker}(\theta) = \{r \in R \mid \theta(r) = 0_s\} \subset R.$$

# THEOREM

## Theorem.

Let  $\theta : R \rightarrow S$  be a ring homomorphism.

1. The image of  $\theta$  is a subring of  $S$ .
2. The kernel of  $\theta$  is an ideal of  $R$ .
3.  $\theta$  is injective if and only if  $\ker \theta = \{0_R\}$ .

## Proof.

Next Practical Session:

1. The image of  $\theta$  is a subring of  $S$ .
2. The kernel of  $\theta$  is an ideal of  $R$ .

Let us then show the third property.

3.  $\theta$  is injective if and only if  $\ker \theta = \{0_R\}$ .



# PROOF OF THEOREM

$\theta$  is injective if and only if  $\ker \theta = \{0_R\}$ .

( $\implies$ ) Suppose  $\theta$  is injective. Let us show  $\ker(\theta) = \{0_R\}$ .

Suppose  $r \in \ker(\theta)$ . This means that  $\theta(r) = 0_S$ .

We know from properties of homomorphisms that  $\theta(0_R) = 0_S$ .

Since  $\theta$  is injective,

$$\theta(r) = 0_S = \theta(0_R) \text{ implies } r = 0_R.$$

This means that the only element of  $\ker(\theta)$  is  $0_R$ .

## PROOF OF THEOREM—PART 2

$\theta$  is injective if and only if  $\ker \theta = \{0_R\}$ .

( $\Leftarrow$ ) Suppose that  $\ker(\theta) = \{0_R\}$ . Let us show that  $\theta$  is injective.

Assume  $\theta(r) = \theta(s)$ . We must show  $r = s$ . Notice that

$$\theta(r - s) = \theta(r + (-s)) = \theta(r) + \theta(-s) = \theta(r) - \theta(s) = 0_S.$$

By definition, this means that

$$r - s \in \ker(\theta).$$

Since  $\ker(\theta) = \{0_R\}$ , we see that  $r - s = 0_R$ . Hence  $r = s$ . □

# THEOREM APPLICATION

## Remark

The previous Theorem can be used to find ideals!

Let us see an example.

## Example

**Last week:** The map  $f : \mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$  given by

$$f(x) = \bar{4}x, \quad \text{for all } x \in \mathbb{Z}/6\mathbb{Z}$$

is a ring homomorphism.

By definition,

$$\text{Ker}(f) = \{z \in \mathbb{Z}/6\mathbb{Z} \mid f(z) = \bar{0}\}.$$

So, we need to find all solutions to  $f(z) = \bar{0}$ . Since  $\mathbb{Z}/6\mathbb{Z}$  has only six elements, we can compute all  $f(z)$ .

## THEOREM APPLICATION - PART 2

### Example

We have

$$\begin{aligned} f(\bar{0}) &= \bar{4} \cdot \bar{0} = \bar{0}, & f(\bar{1}) &= \bar{4} \cdot \bar{1} = \bar{4}, & f(\bar{2}) &= \bar{4} \cdot \bar{2} = \bar{8} = \bar{2}, \\ f(\bar{3}) &= \bar{4} \cdot \bar{3} = \bar{0}, & f(\bar{4}) &= \bar{4} \cdot \bar{4} = \bar{4}, & f(\bar{5}) &= \bar{4} \cdot \bar{5} = \bar{2} \end{aligned}$$

Thus,

$$\text{Ker}(f) = \{\bar{0}, \bar{3}\}.$$

# NEXT LECTURE

## Next time...

- Ideals,
- Principal Ideals,
- Quotient Rings,
- Field of Fractions.